



8)

a)

Source	df	SS	MS	F
Treatment	k-1	SST	$\frac{SST}{k-1}$	$\frac{MST}{MSE}$
Error	N-k	SSE	$\frac{SSE}{N-k}$	
Total	N-1	TSS		

$$CM = \frac{\sum X_{i.}^2}{N} = \frac{840^2}{11} = 64145.45$$

$$TSS = \sum X_{ij}^2 - CM = 1140.55$$

$$SST = \sum \frac{X_{i.}^2}{n_i} - CM = \frac{(13+83+16+68+80)^2}{5} + \frac{(54+74+71)^2}{3} + \frac{(79+95+81)^2}{3} - CM = 28880 + 13200.33 + \dots - CM = 641.883$$

$$SSE = TSS - SST = 1140.55 - 641.883 = 498.67$$

$$MST = \frac{SST}{k-1} = \frac{641.883}{3-1} = 320.94$$

$$F = \frac{MST}{MSE} = \frac{320.94}{62.33} = 5.149$$

$$MSE = \frac{SSE}{N-k} = \frac{498.67}{11-3} = 62.33$$

b)

$$1. H_0: \mu_1 = \mu_2 = \mu_3, H_a: \exists \mu_i \neq \mu_j, i \neq j$$

$$2. \alpha = 0.05$$

$$3. F_{STAT} = \frac{MST}{MSE} \sim F_{k-1, N-k}$$

$$4. F^* = 5.149$$

$$5. R.R. = \{F \mid F > F_{\alpha, k-1, N-k}\} = 4.459$$

$$6. F^* > F_{2, 8, 0.05}$$

The data provide sufficient evidence to indicate a difference

有足夠證據說明三種 Treatment mean 有顯著差異

→ Reject H_0

10.

a. Yes, 有顯著差異. $\therefore p\text{-value} = 0.0251 < 0.05$
 $\Rightarrow \text{Reject } H_0$

b. 99% CI

$$S_p^2 = \frac{3 \times 9.67 + 2 \times 9.33}{4+3-2} = 9.53$$

$$60.50 - 54.67 \pm \underbrace{t_{5, 0.005}}_{403} \sqrt{9.53 \left(\frac{1}{4} + \frac{1}{3} \right)} = \underbrace{(-3.68, 15.34)}_{\text{否}}$$

$\Rightarrow$ A, B mean 沒有 significant differ

c. 99% CI

$$60.5 \pm \underbrace{t_{3, 0.001}}_{5.84} \times \sqrt{\frac{9.67}{4}} = (51.45, 69.55)$$

d. The data satisfy the assumption that they have been selected from Normal population

Since sample is less, and not significant extremely value, ANOVA assumption 大致成立